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In [ ]: # Name - Rohan Pise
        # Rollno. - 01
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In [1]: #Assignments: 9
        #Write a python program to design a Hopfield Network which stores 4 vectors
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import numpy as np
# Define the 4 vectors to be stored
vectors = np.array([[1, 1, -1, -1],
                    [-1, -1, 1, 1],
                    [1, -1, 1, -1],
                    [-1, 1, -1, 1]])

# Calculate the weight matrix
weights = np.zeros((4, 4))
for i in range(4):
    for j in range(4):
        if i == j:
            weights[i, j] = 0
        else:
            weights[i, j] = np.sum(vectors[i] * vectors[j])

# Define the activation function (in this case, a sign function)
def activation(x):
    return np.where(x >= 0, 1, -1)

# Define the Hopfield network function
def hopfield(input_vector, weights):
    output_vector = activation(np.dot(weights, input_vector))
    return output_vector

# Test the Hopfield network with one of the stored vectors as input
input_vector = vectors[0]
output_vector = hopfield(input_vector, weights)
print("Input vector:")
print(input_vector)
print("Output vector:")
print(output_vector)
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Input vector:
[ 1  1 -1 -1]
Output vector:
[-1 -1  1  1]
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In [ ]:
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